The Riemann Sums

F18CE Week 7

Anti-derivative

Definition. Let $F, f : \mathbb{R} \to \mathbb{R}$ be functions.

Then F is said to be an anti-derivative of f if

$$F' = f\,.$$

Example of anti-derivative

$F(x) = \frac{1}{2}x^2$ is an anti-derivative of $f(x) = x$ because $F'(x) = f(x)$,

but $G(x) = \frac{1}{2}x^2 + 1$ is also an anti-derivative of $f(x) = x$ because

$G'(x) = f(x)$. So, the anti-derivative of a function may not be unique.

Remark

Thus, all the antiderivatives of $f(x) = x$ can be obtained by changing the value of C in $F(x) = \frac{1}{2}x^2 + C$, where C is an arbitrary constant known as the constant of integration

Anti-derivative vs area under the curve

Newton, Leibniz, Fermat, and the other founders of calculus knew there exists some relationship between antiderivatives and areas (under the curve of functions).

However, if we only consider the computation of area under the curve for differentiable functions, then this approach is ultimately unsatisfying from the point of view of analysis because it results in a very limited number of functions that can be integrated.

This means that any function with a jump discontinuity cannot be a derivative. If we want to define integration via anti-differentiation, then we must accept the consequence that a function as simple as

$$h(x) = \begin{cases} 1, & 0 \leq x < 1 \\ 2, & 1 \leq x < 2 \end{cases}$$

is not integrable on the interval $[0, 2]$.

Concept of integration

The integration, as we study now, is the *area* under the curve of a function $f(x)$ with the x-axis. Note that "area" here comes with the understanding that areas below the x-axis are assigned a negative value.

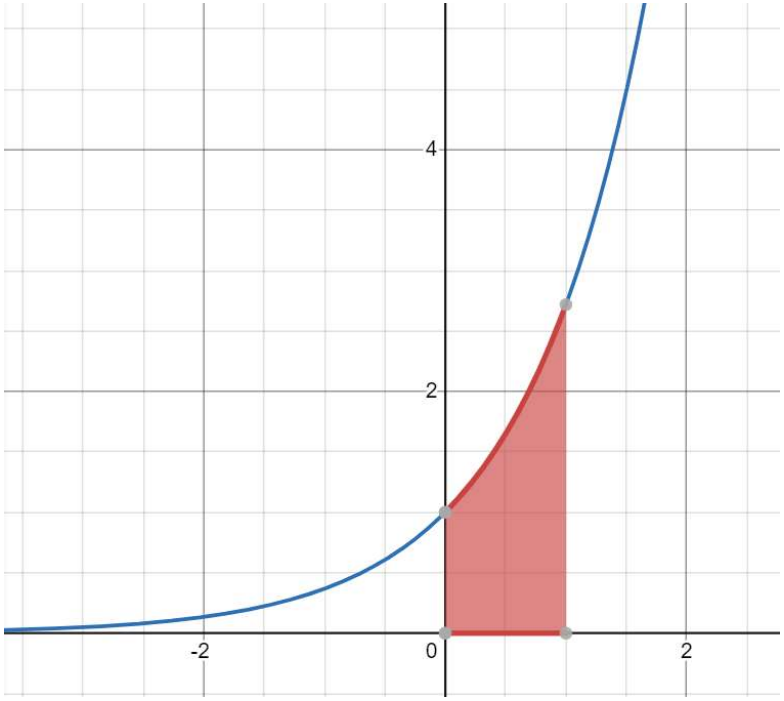
We introduce the Riemann sum to *help* evaluate the integration but ultimately, we shall introduce the Fundamental theorem of calculus to link the antiderivative with integration.

An introductory example

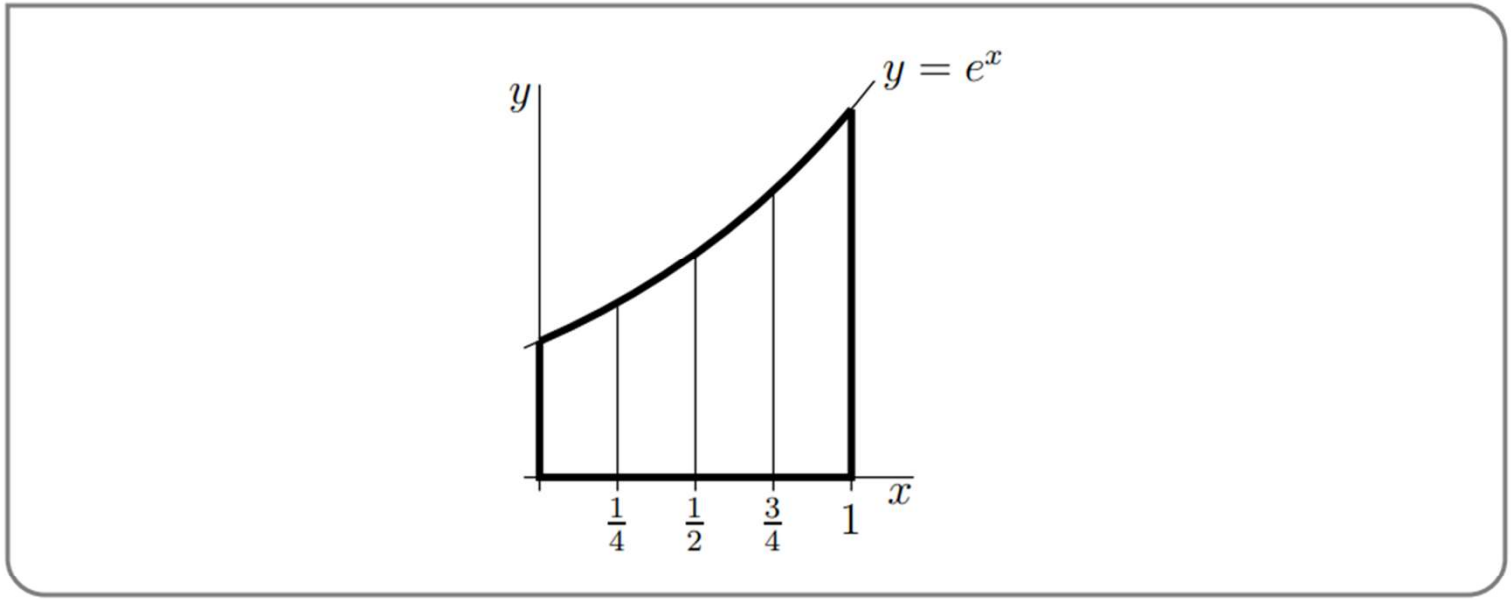
Suppose we want to find the area under the curve

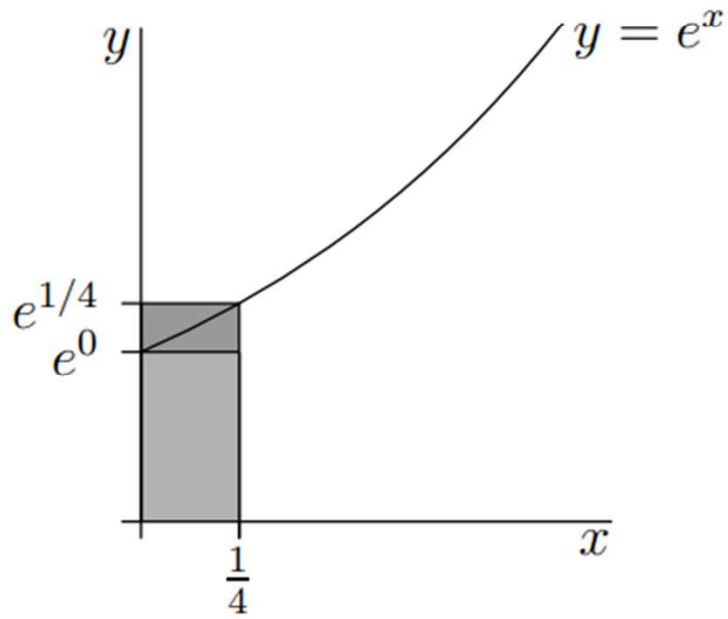
$$y = e^x \text{ for } 0 \le x \le 1\,,$$

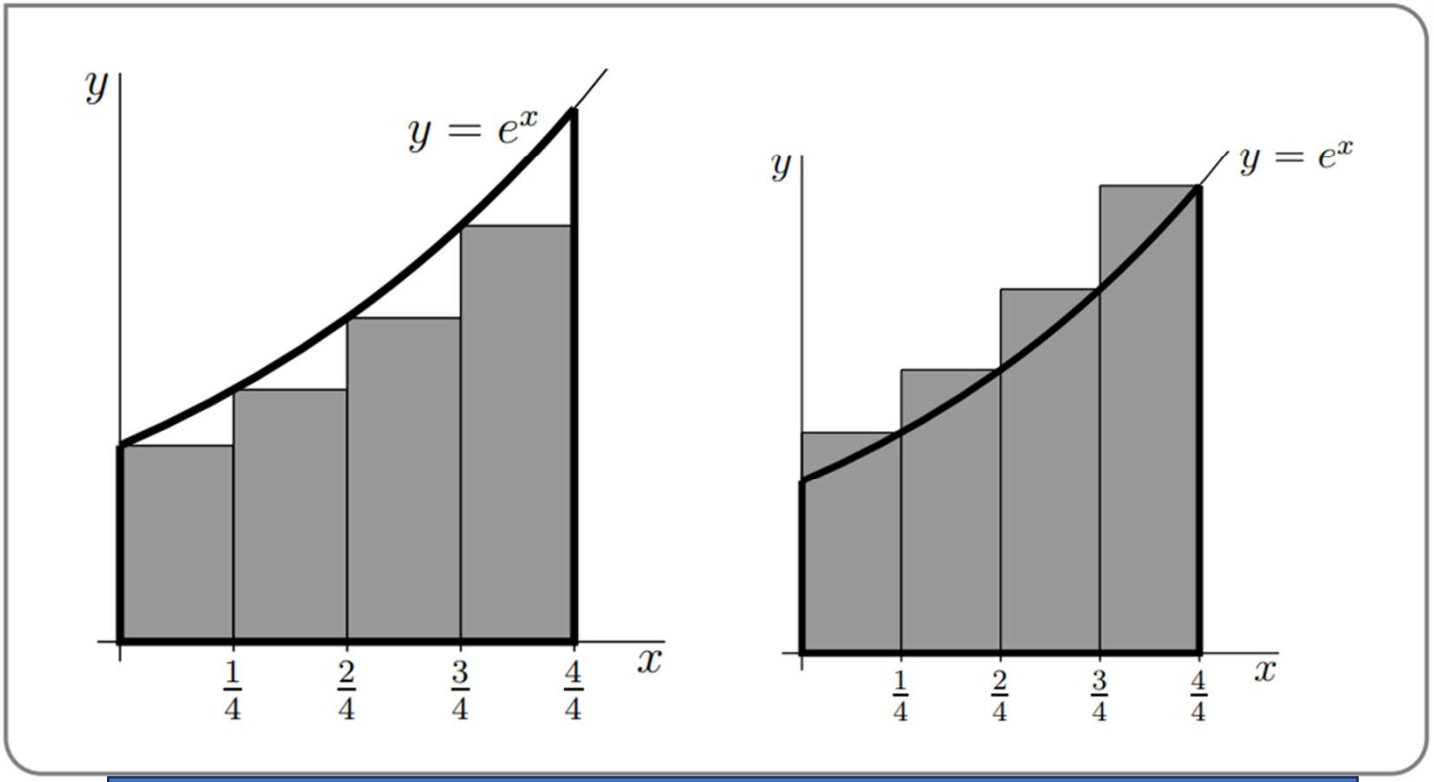
i.e., the area of $D = \left\{(x,y) : 0 \le y \le e^x, 0 \le x \le 1\right\}$.



A way to find (estimate) the area:




y
$y = e^x$
$e^{1/4}$
e^0
x
$\frac{1}{4}$



L (Lower Riemann sum)

U (Upper Riemann sum)

Area approximated is $\frac{1}{4}\left(e^0 + e^{1/4} + e^{1/2} + e^{3/4}\right)$

we call this as the lower Riemann sum to $\int_0^1 e^x dx$,

written $L = \frac{1}{4}\left(e^0 + e^{1/4} + e^{1/2} + e^{3/4}\right)$.

On the other hand, we have the upper Riemann sum to $\int_0^1 e^x dx$, written $U = \frac{1}{4}\left(e^{1/4} + e^{1/2} + e^{3/4} + e\right)$.

Note that $L \leq \int_0^1 e^x dx \leq U$.

In general

Suppose we partition $[0, 1]$ into n subintervals with length of $\frac{1}{n}$ each; (and later take $n \to \infty$). Then we have

$$L = \frac{1}{n}\left(e^0 + e^{1/n} + e^{2/n} + \cdots + e^{(n-1)/n}\right) = \frac{1-e}{n(1-e^{1/n})}$$

$$U = \frac{1}{n}\left(e^{1/n} + e^{2/n} + \cdots + e^{(n-1)/n} + e\right) = \frac{e^{1/n}(1-e)}{n(1-e^{1/n})}$$

Take $n \to \infty$, we have

$$\lim_{n\to\infty} L = \lim_{n\to\infty} \frac{1-e}{n(1-e^{1/n})} = (1-e) \lim_{x\to 0^+} \frac{x}{1-e^x} = e-1$$

(substitute $x = \frac{1}{n}$ and so $x \to 0^+$ as $n \to \infty$)

and $\lim_{n\to\infty} U = \lim_{n\to\infty} \frac{e^{1/n}(1-e)}{n(1-e^{1/n})} = e-1$.

Hence, $\int_0^1 e^x dx = e-1$.

Riemann sum – Introduction

Let $f : [a,b] \to \mathbb{R}$ be a bounded function. The aim of this chapter is to Define the integral of f over $[a,b]$.

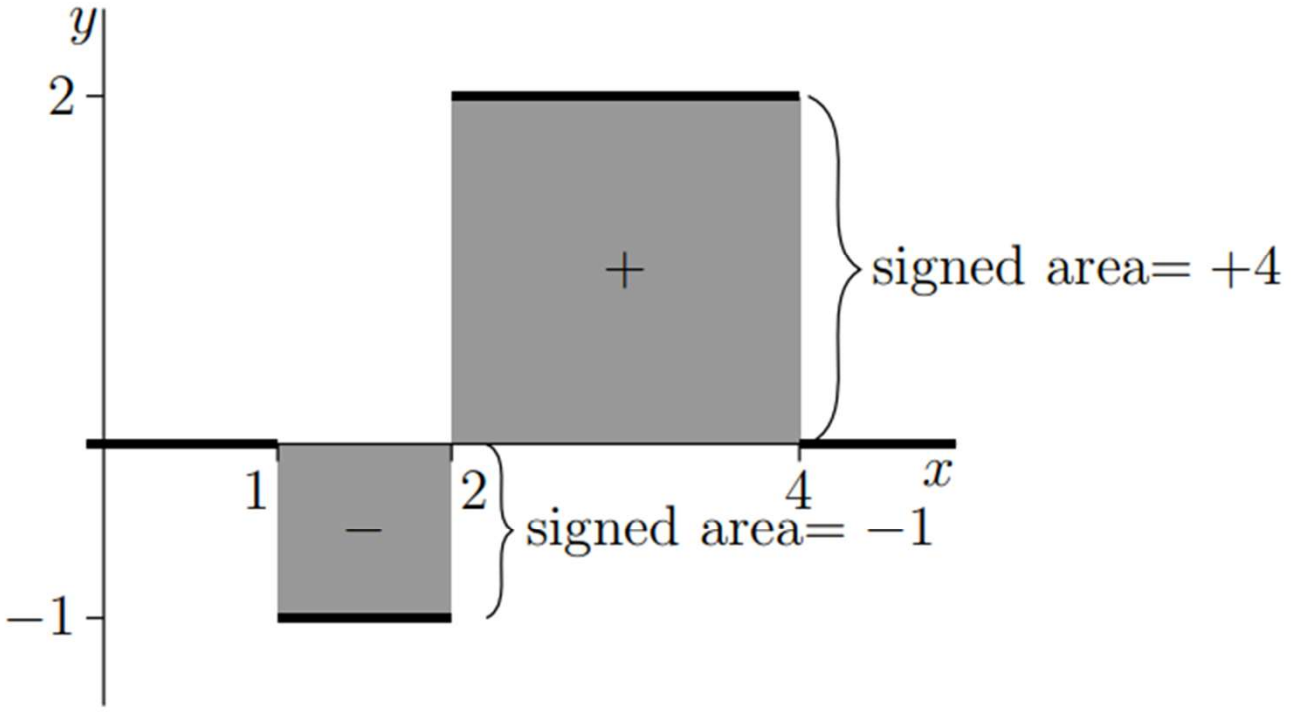
We consider the area under the curve of f (or more precisely, the area between the graph of f and the x–axis).

Also, we define the area as negative if the graph of f is below the x–axis.

Signed area

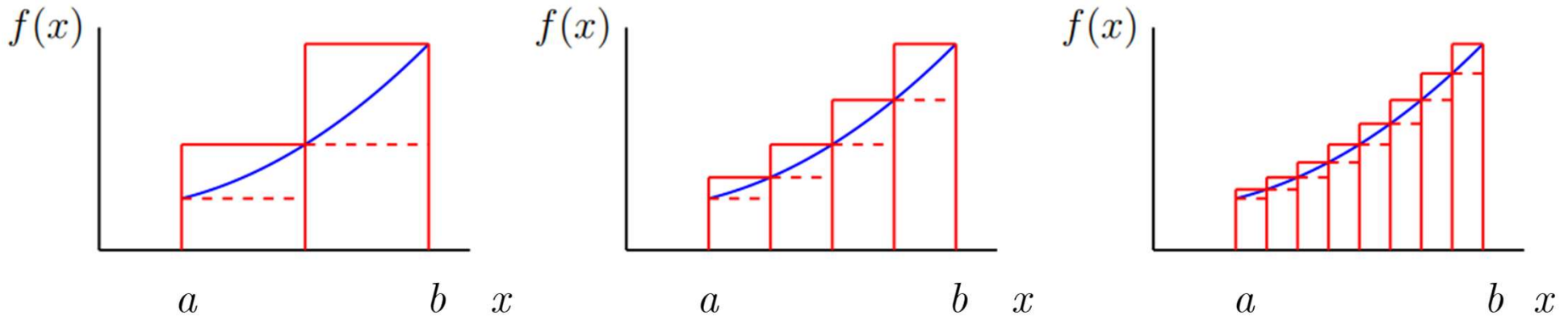
Recall if $f(x) < 0$ for some part in $a \leq x \leq b$. Then we shall take the area as negative.

(We call this area as "signed area"). For example, let $f(x) = \begin{cases} -1, & 1 \leq x < 2 \\ 2, & 2 < x \leq 4 \\ 0, & \text{otherwise} \end{cases}$. Then


y
2
−1
1
2
4
x
+
−
signed area= +4
signed area= −1

Upper vs lower Riemann Sum

For any function f, we can partition the graph to sets of rectangles as below.



Lower Riemann sum is defined as the smallest possible area from the rectangles, and Upper Riemann sum is the largest possible area from the rectangles. When the function is partition into infinitely many rectangles, then intuitively we see that the exact value for the area is obtained.

Left, middle and right Riemann sum

We can also define the left; middle; and right Riemann sum by taking the area measuring the height from the most-left; the middle; and the most-right of each sub-interval.

Referring to the previous example, the left Riemann sum = lower Riemann sum, and the right Riemann sum = upper Riemann sum (because the function is increasing).

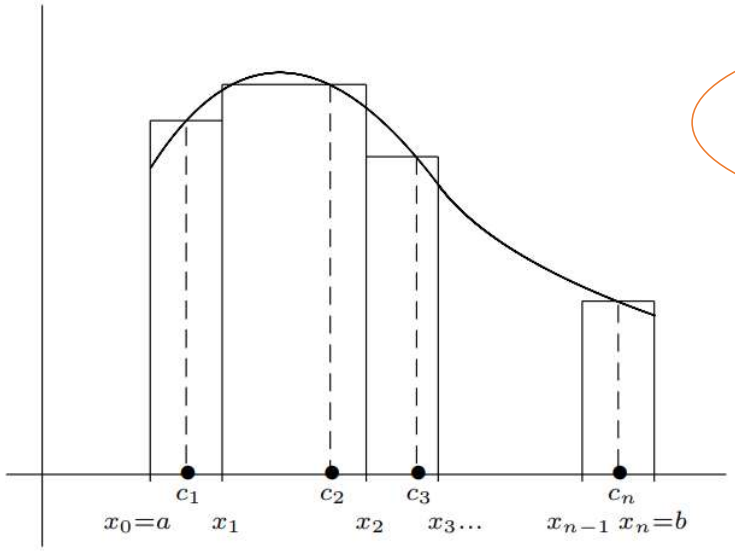
Can you find the middle Riemann sum for the previous example?

Anywhere is fine?

Starting with a function f on $[a,\ b]$, we partition the domain into small subintervals. On each subinterval $\left[x_{k-1}, x_k\right]$, we pick some point $c_k \in \left[x_{k-1}, x_k\right]$ and use the y-value $f(c_k)$ as an approximation for f on $\left[x_{k-1}, x_k\right]$.

Graphically speaking, the result is a row of thin rectangles constructed to approximate the area between f and the x-axis.

The area of each rectangle is $(x_k - x_{k-1})f(c_k)$, and so the total area of all of the rectangles is given by the Riemann sum $\sum_{k=1}^{n}(x_k - x_{k-1})f(c_k)$.



$c_k \in (x_{k-1}, x_k)$ can be arbitrary

Partitions (Definition 6.1)

Let P be a set of points, $P = \{x_0, x_1, \ldots, x_k\}$ such that

$$a = x_0 < x_1 < \cdots < x_k = b\,.$$

Then P is said to be a partition of $[a,b]$.



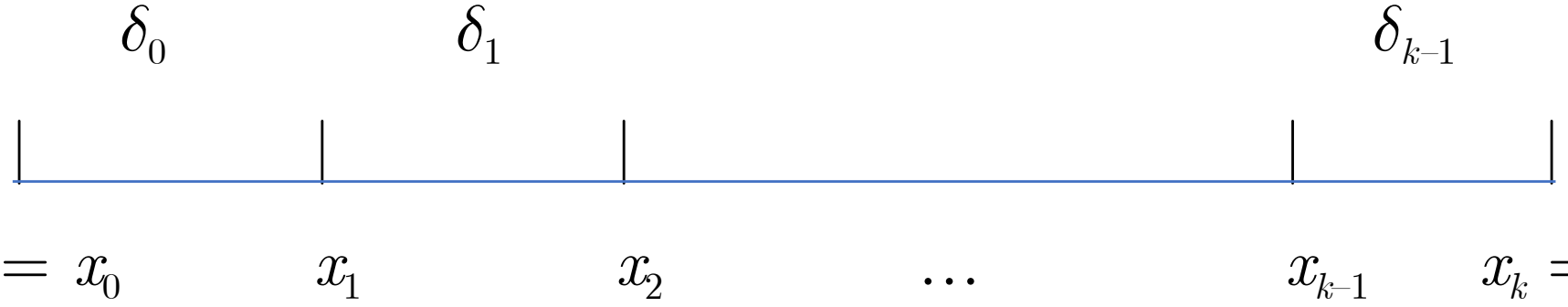
$a = x_0 \quad x_1 \quad x_2 \qquad\qquad x_k = b$

Maximum width of subintervals

For a partition P of $[a,b]$, define

$$\delta_r := x_{r+1} - x_r \quad \forall r = 0, 1, \ldots, k-1; \text{ and}$$

$$\delta_{\max}(P) := \max\{\delta_0, \ldots, \delta_{k-1}\}.$$

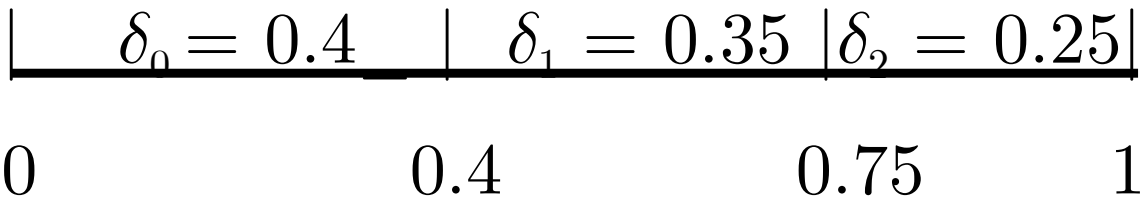


$\delta_{\max}(P)$ is the maximum (longest) width of all the subintervals.

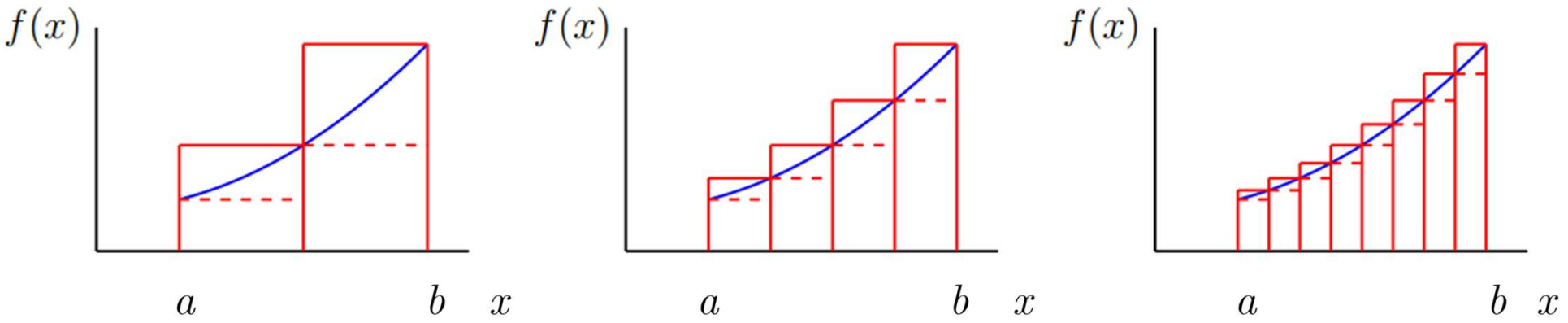
Example 6.2

Let $P = \{0, 0.4, 0.75, 1\}$ be a partition of $[0, 1]$.

Then $\delta_0 = 0.4$, $\delta_1 = 0.35$, $\delta_2 = 0.25$.



Remark



A partition P splits the interval $[a,b]$ into k subintervals $[x_r, x_{r+1}]$, $r = 0, 1, \ldots, k-1$; with width δ_r, and these subintervals will be the bases of the rectangles (this is the first step in the above process of calculating the area).

Remark

The subintervals need not all have the same width.

The sum of the width of all subintervals equals to the total width of $[a,b]$,

which is $\sum_{r=0}^{k-1} \delta_r = \sum_{r=0}^{k-1} (x_{r+1} - x_r) = b - a$.

Riemann Sum (Definition 6.3)

A *Riemann sum* with respect to the partition P is any sum of the form

$$S(P,f) = \sum_{r=0}^{k-1} f(x_r^*)\delta_r \ ,$$

where $r = 0, 1, \ldots, k-1$, and $x_r^* \in \left[x_r, x_{r+1}\right]$.

Lower and upper Riemann sums (Definition 6.4)

The lower Riemann sum $L(P,f)$ and the upper Riemann sum $U(P,f)$ of f with respect to the partition P are defined as

$$L(P,f)=\sum_{r=0}^{k-1} I_r(f)\delta_r\ , \qquad U(P,f)=\sum_{r=0}^{k-1} S_r(f)\delta_r \qquad \forall r = 0,\ 1,\ \ldots,\ k-1;$$

where $I_r(f)=\inf\{f(x): x\in[x_r,x_{r+1}]\}$ and $S_r(f)=\sup\{f(x): x\in[x_r,x_{r+1}]\}$.

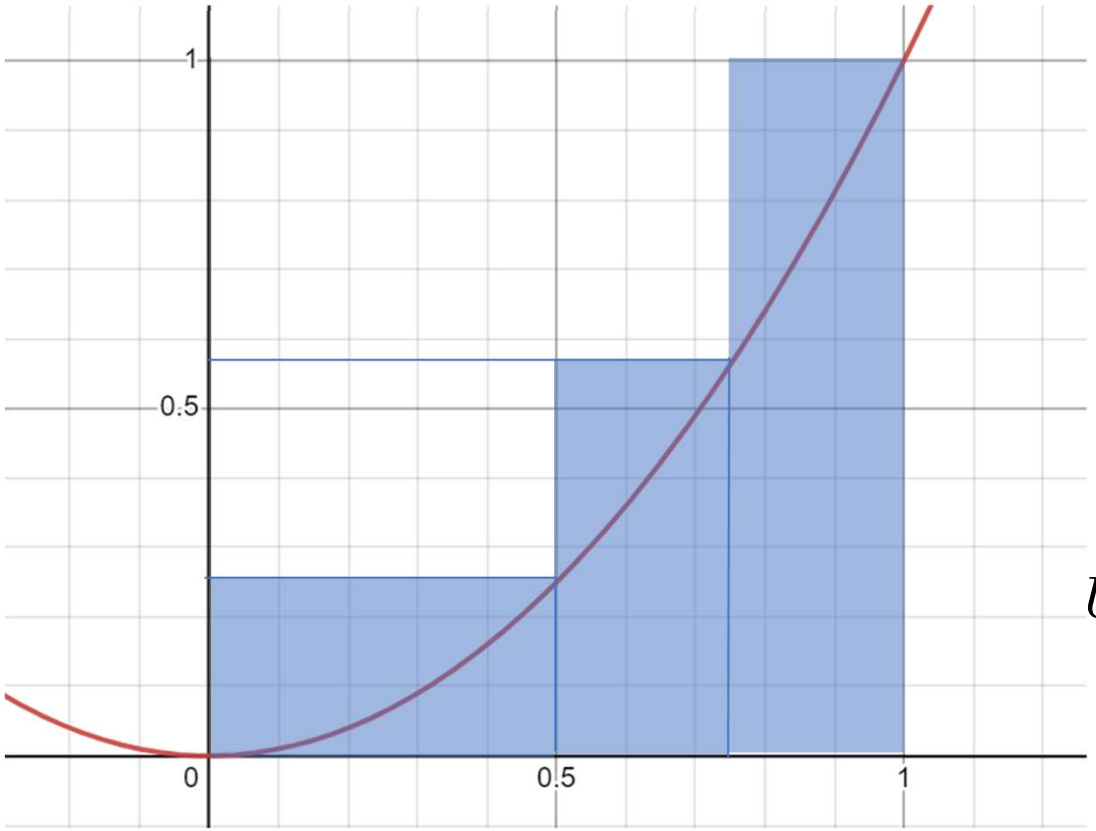
Remark

To make it easier, if f is continuous, then $I_r(f)$ and $S_r(f)$ are the minimum and maximum of f in the subinterval $[x_r, x_{r+1}]$, respectively.

$L(P, f)$ and $U(P, f)$ are areas under the set of lower and upper rectangles, respectively.

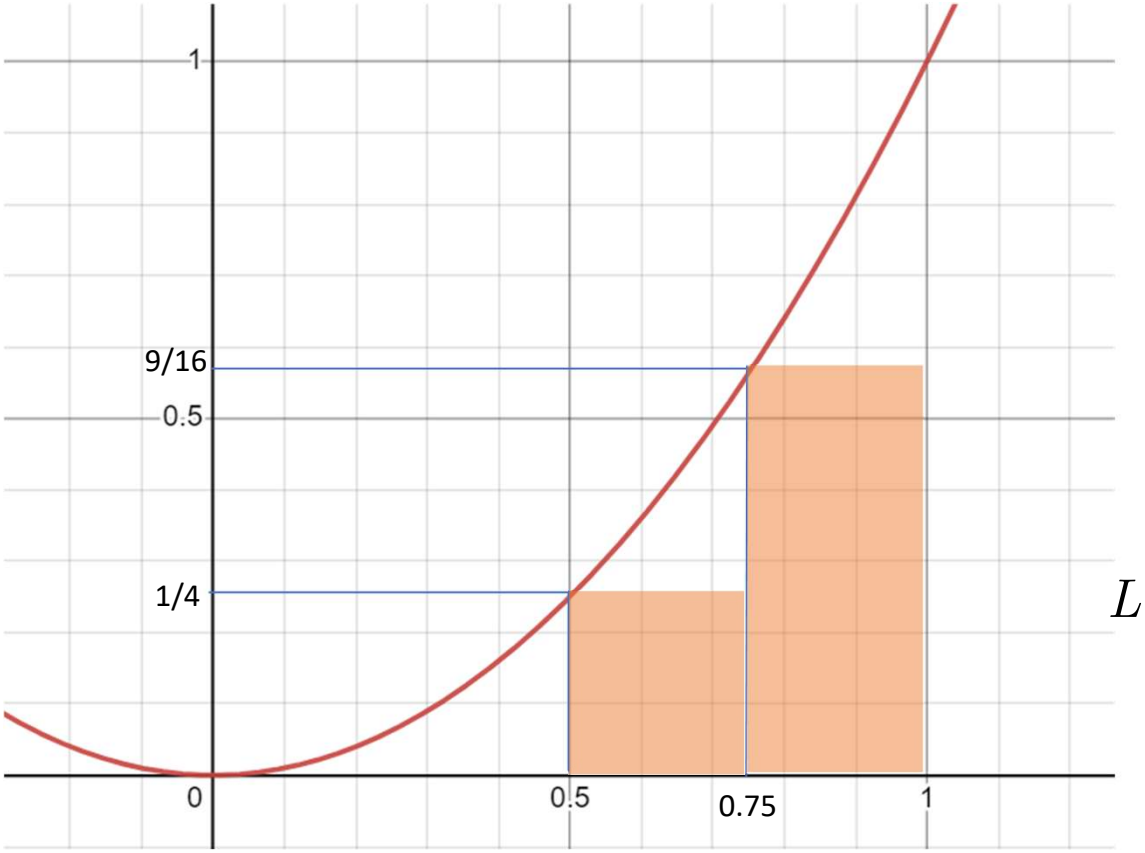
Example 6.5

Let $f(x) = x^2$ and let $P = \{0, \frac{1}{2}, \frac{3}{4}, 1\}$.



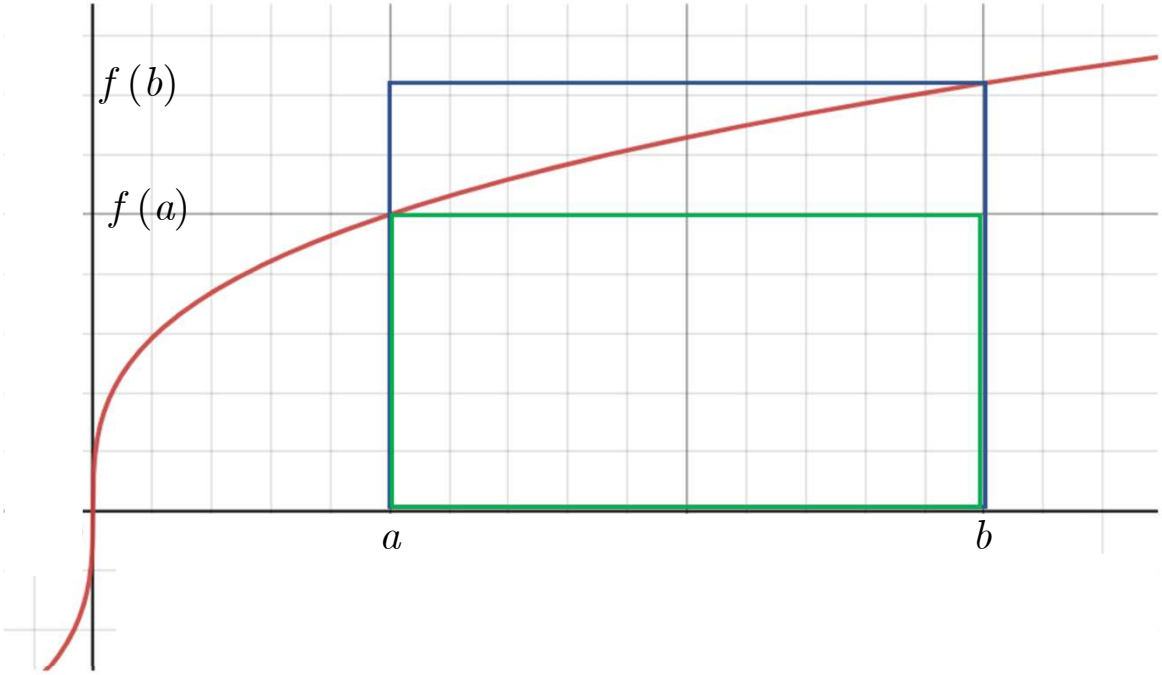
$$U(P, f) = \frac{1}{2} \times \frac{1}{4} + \frac{1}{4} \times \frac{9}{16} + \frac{1}{4} \times 1 = \frac{33}{64}$$

Let $f(x) = x^2$ and let $P = \{0, \frac{1}{2}, \frac{3}{4}, 1\}$.



$$L(P, f) = 0 + \frac{1}{4} \times \frac{1}{4} + \frac{1}{4} \times \frac{9}{16} = \frac{13}{64}$$

Suppose partition of $[a,b]$ is defined as $T_0 := \{a,\ b\}$. Then $L(T_0,f) \leq U(T_0,f)$.



$$L(T_0,f) = (b-a)f(a)$$
$$U(T_0,f) = (b-a)f(b)$$

It is clear that $f(b) > f(a) \Rightarrow U(T_0,f) > L(T_0,f)$

Lemma 6.6

For any partition P of $[a,b]$, we have

$$L(T_0, f) \leq L(P, f) \leq U(P, f) \leq U(T_0, f).$$

Refinement

Next, suppose that we have two partitions P and R, such that $P \subset R$, that is, all the points in P are also in R.

In this case we say that R is a *refinement* of P.

We refine a partition P by adding points to it to give the new partition R, so that either (i) all the gaps have the same size, or (ii) the gaps get smaller.

Lemma 6.7

Let P and R be partitions of $[a,b]$ such that R is a refinement of P. Then

$$L(P,f) \leq L(R,f) \qquad \text{and} \qquad U(R,f) \leq U(P,f).$$

Theorem 6.8

The following theorem deals with any two partitions P and Q, neither of which need be a refinement of the other one.

Let P and Q be any partitions of $[a,b]$. Then

$$L(P,f) \leq U(Q,f).$$

Remark

Of course, this is true also: $L(Q, f) \leq U(P, f)$.

In general, any lower Riemann sum of a partition is less than or equal to any upper Riemann sum of another partition.